Predictive Analytics Poster Rubric (Poster-Only Assessment)

#	Criterion	Points	Exemplary (Full Points)	Competent (Partial Points)	Needs Improvement (Minimal Points)
1	Prediction Goal(s)	5	5 pts: Clear, specific predictive objective(s); response type (regression/classification) explicitly identified and justified; strong business/research rationale.	3–4 pts: Goal stated; response type identified but weakly justified; limited link to context.	0–2 pts: Goal unclear/missing; response type unspecified; no rationale.
2	Dataset Overview	8	7–8 pts: Reports #obs/#features; types for key features; response variable identified/typed; data provenance/window stated.	4–6 pts: Size and response identified; partial feature typing; minimal provenance.	0–3 pts: Missing size/response ID; little/no feature typing or context.
3	Exploratory Data Analysis (EDA)	12	10–12 pts: Appropriate summaries (response + key predictors); ≥2 relevant plots; interprets skew/outliers/imbalance/seasonality; connects insights to modeling choices and metric selection.	6–9 pts: Basic stats and ≥1 plot; some interpretation but limited linkage to modeling.	0–5 pts: Incomplete stats; weak/irrelevant visuals; minimal/no interpretation.
4	Data Preparation (Feature Engineering, Missing Data, etc.)	18	15–18 pts: ≥1 non-trivial new feature (interactions, lags/rolling stats, domain-driven ratios, text-derived signals) with clear rationale; quantifies missingness by feature; justified strategy (drop/impute/advanced) with leakage-safe implementation; demonstrates impact via ablation/CV deltas or sensitivity to imputation choices.	9–14 pts: New feature with partial rationale; identifies missingness and applies a reasonable method; limited evidence of impact or sensitivity analysis.	0–8 pts: No new/trivial feature; missingness unreported or mishandled; unjustified/inappropriate method; no evidence of value.
5	Baseline Model (Choice, CV, Interpretation)	17	14–17 pts: Appropriate simple baseline (e.g., Linear/Logistic Regression); pipeline is leakage-safe; k=5/10 CV with correct metric (RMSE/MAE/Accuracy/F1/AUC); tracks mean/variance; summarizes implications and limitations; proposes concrete next steps.	8–13 pts: Reasonable model; CV present with partial integration/details; metric adequate; interpretation limited or generic.	0–7 pts: Inappropriate/unspecified baseline; no/incorrect CV; unclear metric; raw numbers only; no reflection.

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6	More Complex Model (Choice, Tuning, Paired Comparison)	25	22–25 pts: Model aligned to data/goal (RF/GBM/XGBoost/SVM/regularized GLM/NN); hyperparameter search (grid/random/Bayesian) inside proper CV; leakage controls (scaling/encoding inside folds); compared with the baseline on identical folds (same splitter and seed) using per-fold paired differences (positive favors the complex model) against a margin stated before the results; mean difference reported with an approximate 95% interval corrected for fold overlap: mean ± t × SD(differences) × √(1/k + 1/(k−1)); verdict names the outcome: worthwhile gain (interval entirely above the margin), worthwhile loss (entirely below minus the margin), practically equivalent (entirely inside ± the margin: the simpler model wins), or unresolved (anything else: the simpler model stays as a stated policy, not called equivalent); practical interpretation (effect sizes, trade-offs), plus error/confusion analysis where relevant.	13–21 pts: Plausible model; some tuning with partial documentation; CV present but thin; compares with the baseline on CV means or side-by-side intervals rather than paired per-fold differences; judges a winner or a tie by whether intervals overlap or by fold signs alone; calls the models equivalent when the interval extends past the margin; or states no margin; limited practical takeaways.	0–12 pts: Misaligned/unspecified model; superficial/absent tuning; incorrect CV; opaque selection; no clear comparison or interpretation.
7	Report Quality & Visual Communication	10	9–10 pts: Clear, scannable poster; logical flow (Intro → Methods → Results → Conclusions); legible, labeled figures/tables (titles, axes, units, legends); consistent typography/layout; code refs/appendix where appropriate; minimal grammar issues.	6–8 pts: Generally readable; minor layout/labeling gaps; some figures under-explained or styling inconsistencies.	0–5 pts: Disorganized; illegible/unlabeled visuals; weak linkage between text and graphics; notable grammar/formatting issues.
8	Sources, Acknowledgments, AI Use & Data Privacy	5	5 pts: Data source cited (provider, URL or DOI, version or access date); references support the business-problem claims and the main software is credited (e.g., scikit-learn, pandas); Acknowledgments section names the Mitch Daniels School of Business, Prof. Moreira, the TAs, and other contributors; AI Use Disclosure names each AI tool and what it was used for and states that the team verified the output (or states that no AI tools were used); only aggregated or anonymized data and examples.	3–4 pts: AI Use Disclosure present but vague (names tools without saying what they did), or one other element missing or incomplete (e.g., acknowledgments omit the TAs; data source cited without references or tools). No privacy issue.	0–2 pts: No AI Use Disclosure, or two or more other elements missing. Any personally identifiable or NDA-restricted data on the poster scores 0 here and must be removed before printing.

Total: 100 points. This rubric grades M09 (poster first draft), M10 (final poster), and the Instructor/TA evaluation after the conference.

Criterion 6 at M09: missing paired differences, margin, or corrected interval get feedback, not a deduction, on the first draft; all three are scored from M10.

Fall 2026: a comparison done as taught in notebook 08 (descriptive interval; verdict from its three outcomes: every fold favors one model and the gap clears the margin, the sign changes across folds, or the same sign under the margin) also earns full credit when the outcome is applied correctly.

AI use: per the syllabus AI Policy, every poster carries an AI Use Disclosure (the template has a section for it). Errors in AI-assisted work are the team's responsibility.

Honors: a labeled honors section, where present, is graded only under the Honors Research Extension and is not scored under this rubric.